

IE684: Operations Research Lab

Binary Classification & Margin Optimization

Motupalli Hemanth Kumar
Roll No: 25M1512

February 13, 2026

Exercise 1: Loss Functions Behavior

The objective of binary classification is to learn a rule $h(x) = \text{sign}(w^T x)$. The behavior of different loss functions relative to the margin $z = yw^T x$ is summarized below:

- **Hinge Loss (G_h):** $G_h(z) = \max\{0, 1 - z\}$. It yields 0 for $z \geq 1$ (margin met) and applies a linear penalty for $z < 1$. It is non-differentiable at the exact margin boundary ($z = 1$).
- **Squared Hinge Loss (G_{sh}):** $G_{sh}(z) = (\max\{0, 1 - z\})^2$. It applies a quadratic penalty for margin violations, making it smoother and differentiable everywhere, but significantly harsher on extreme outliers.
- **Logistic Loss (G_l):** $G_l(z) = \log(1 + \exp(-z))$. It is strictly greater than 0 everywhere, smooth, and convex, behaving linearly for large negative values of z .

Exercise 2: Data Preparation

The standard Iris dataset was loaded via `scikit-learn`. The 3-class target vector was converted into a binary classification problem, where Class 1 was assigned +1 and all other classes were assigned -1. The dataset of $n = 150$ samples was randomly shuffled and split into an 80% training set and a 20% testing set to validate generalizability.

Exercise 3: Optimization Algorithm

3.1 Mathematical Formulation

The total objective function is expressed as a sum over n terms by distributing the L_2 regularization term ($\frac{\lambda}{2}\|w\|_2^2$) across all data points:

$$f_i(w) = \frac{\lambda}{2n}\|w\|_2^2 + \frac{1}{n}\max\{0, 1 - y_i w^T x_i\}$$

The sub-gradient $g_i(w) = \nabla_w f_i(w)$ depends on the margin violation:

- **Violation ($1 - y_i w^T x_i > 0$):** $g_i(w) = \frac{\lambda}{n}w - \frac{1}{n}y_i x_i$
- **Correct Classification ($1 - y_i w^T x_i \leq 0$):** $g_i(w) = \frac{\lambda}{n}w$

3.2 Regularization Tuning (λ)

The Stochastic Subgradient Descent (SGD) algorithm was tested over 1000 epochs with $\lambda \in \{10^{-3}, 10^{-2}, 0.1, 1, 10\}$.

- $\lambda = 10^{-3}$: Led to overfitting (high training accuracy, low test accuracy).
- $\lambda = 10$: Led to underfitting (poor performance across both sets).
- $\lambda = 0.01$: Emerged as the optimal regularization parameter, consistently achieving the highest test-set accuracy by effectively managing the bias-variance tradeoff.

3.3 Early Stopping Criterion

An alternative stopping criterion was implemented to monitor objective convergence. By halting the algorithm when the absolute difference in the objective function between consecutive epochs fell below a tolerance of 10^{-5} , the model converged efficiently (often between 50 and 200 epochs) rather than requiring the full 1000 iterations.

3.4 Comparison of Loss Functions

- **Convergence:** Logistic loss provided the smoothest convergence curve due to its strict differentiability. Hinge loss displayed a more "jagged" update trajectory due to the usage of sub-gradients at the margin boundary.
- **Performance:** Despite the mathematical and outlier-sensitivity differences between Hinge, Squared Hinge, and Logistic functions, all three achieved a high accuracy ($> 95\%$). This is primarily because the classes within the binary-transformed Iris dataset are easily separable, allowing multiple decision boundaries to perform optimally.